

AUTOSAR High-Level Design (HLD) Specification

Domain: Powertrain & Chassis Control Domain | Release 1.0 (Baseline)

Document ID: DOC-AUTOSAR-PT-V1	Target ECU: TriCore TC397x (Domain Controller)
Standard: AUTOSAR Classic Platform 4.4.0	Classification: ASIL-D / ISO 26262
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1. System Architecture Overview

This High-Level Design (HLD) document specifies the software architecture for the Powertrain and Chassis Domain Controller in compliance with AUTOSAR Classic Release 4.4. The architecture defines application software components (SWCs), sender-receiver and client-server interfaces, port prototypes, signal mapping, and data type definitions necessary for distributed torque control and regenerative braking coordination.

2. Software Component: BrakeControlSWC

Component Type: Application Software Component (ApplicationSWC)
Execution Periodicity: 10ms (Cyclic Runnable: BrakeControl_Runnable_10ms)
Safety Level: ASIL-D
Description: Responsible for calculating total friction and regenerative braking demand based on driver brake pedal input, wheel speed sensor feedback, and vehicle dynamic stability.

Ports & Interface Definitions for BrakeControlSWC:

Port Name	Port Type	Interface Name	Signal / Element	Data Type	Target / Provider
RPort_VehicleSpeed	R-Port (Require)	If_VehicleSpeed	VehicleSpeed_kph	float32	WheelSpeedSensorSWC
PPort_BrakeTorqueRequest	P-Port (Provide)	If_BrakeTorque	BrakeTorque_Nm	uint16	EngineManagerSWC
RPort_SteeringAngle	R-Port (Require)	If_SteeringAngle	SteeringAngle_deg	float32	SteeringAngleSensorSWC

3. Software Component: EngineManagerSWC

Component Type: Application Software Component (ApplicationSWC)

Execution Periodicity: 5ms (Cyclic Runnable: EngineManager_Step_5ms)

Safety Level: ASIL-B

Description: Controls primary combustion and e-motor drive torque synthesis, throttle response, and coordinates torque reduction commands received from chassis safety components during stability interventions.

Ports & Interface Definitions for EngineManagerSWC:

Port Name	Port Type	Interface Name	Signal / Element	Data Type	Target / Provider
RPort_BrakeTorque	R-Port (Require)	If_BrakeTorque	BrakeTorque_Nm	uint16	BrakeControlSWC
PPort_EngineSpeed	P-Port (Provide)	If_EngineSpeed	EngineSpeed_rpm	uint16	TransmissionControlSWC
RPort_ThrottleDemand	R-Port (Require)	If_ThrottleDemand	ThrottlePercent	uint8	PedalInterfaceSWC

4. Software Component: BatteryManagementSWC

Component Type: Sensor/Actuator Software Component (SensorActuatorSWC)

Execution Periodicity: 50ms (Cyclic Runnable: BMS_Main_50ms)

Safety Level: ASIL-C

Description: Monitors high-voltage battery pack State of Charge (SOC), State of Health (SOH), pack terminal voltage, and maximum discharge current limits.

Port Name	Port Type	Interface Name	Signal / Element	Data Type	Target / Provider
PPort_BatterySOC	P-Port (Provide)	If_BatteryStatus	BatterySOC_pct	uint8	TransmissionControlSWC
PPort_BatteryCurrent	P-Port (Provide)	If_BatteryCurrent	Current_A	uint16	TransmissionControlSWC

5. Software Component: TransmissionControlSWC

Component Type: Application Software Component (ApplicationSWC)

Execution Periodicity: 20ms (Cyclic Runnable: TCU_Control_20ms)

Safety Level: ASIL-C

Description: Controls automated dual-clutch transmission gear selection, clutch engagement schedules, and hybrid drive mode coupling based on engine speed and battery charge thresholds.

Port Name	Port Type	Interface Name	Signal / Element	Data Type	Target / Provider
RPort_EngineSpeed	R-Port (Require)	If_EngineSpeed	EngineSpeed_rpm	uint16	EngineManagerSWC
RPort_BatteryCurrent	R-Port (Require)	If_BatteryCurrent	Current_A	float32	BatteryManagementSWC
PPort_GearPosition	P-Port (Provide)	If_GearStatus	CurrentGear	uint8	InstrumentClusterSWC

6. Functional Flows & End-to-End Traces

Flow 1: Regenerative Braking Coordination

- BrakeControlSWC receives wheel speeds via RPort_VehicleSpeed.
- BrakeControlSWC computes negative torque command and emits PPort_BrakeTorqueRequest.
- EngineManagerSWC receives torque demand and reduces internal combustion output.
- TransmissionControlSWC commands optimal gear ratio for regenerative recovery.

Flow 2: High Voltage Battery Overcurrent Protection

- BatteryManagementSWC samples pack current via PPort_BatteryCurrent.
- TransmissionControlSWC reads current via RPort_BatteryCurrent to modulate hybrid motor load.

7. Standard Data Type Definitions

Data Type	Base Type	Unit	Resolution / Range
VehicleSpeed_kph	float32	km/h	0.01 resolution (0.0 to 350.0)
BrakeTorque_Nm	uint16	Nm	1.0 resolution (0 to 65535)
EngineSpeed_rpm	uint16	rpm	1.0 resolution (0 to 12000)
Current_A	uint16	Amperes	0.1 resolution (-500 to +500)
BatterySOC_pct	uint8	%	1% (0 to 100)